

# ECE 447

## Lesson 24

### Pulse Shaping & Intersymbol Interference (ISI)



UNITED STATES  
AIR FORCE  
ACADEMY

# SCHEDULE AND ADMIN

- [Schedule](#)
- Admin
  - **HW5.** Was due Lesson 23 (13 Oct).
  - **Lab 5.** Due tonight (Lesson 24, 15 Oct) to Gradescope.
  - **HW6.** Assigned today. Due Lesson 28 (Tue, 27 Oct) to Gradescope.
  - Problem 4 of HW6 uses  $M$ -ary signaling from Lesson 26 – start with Problems 1-3.

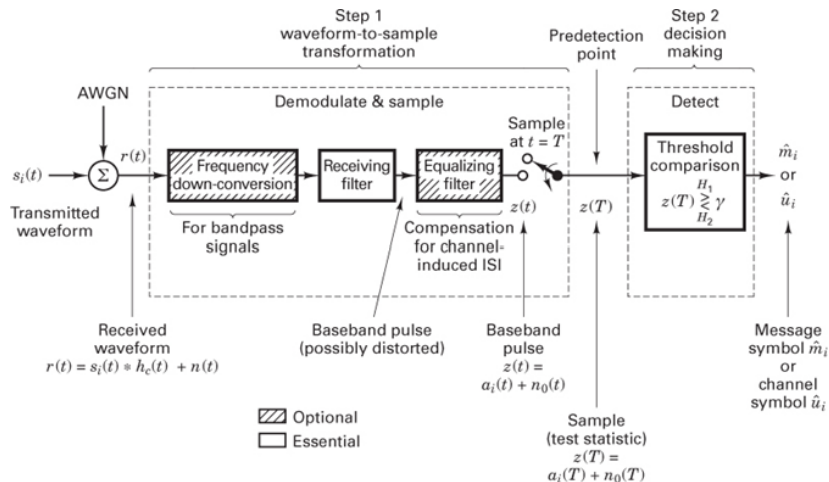
# REVIEW

- Line codes - set of symbols  $a_k$  that determine the amplitude of the baseband pulses,  $p(t)$
- Sum of scaled pulses transmitted at  $R_b = 1/T_b$  pulses per second:  $y(t) = \sum_k a_k p(t - kT_b)$
- Desired properties:
  - Low bandwidth
  - Power efficiency
  - Error detection/correction capability
  - No DC content
  - Timing info (self-clocking)
  - Transparency
- PSD of  $y(t)$  controlled both by line code and pulse shape
- Time-limited signal (e.g.,  $p(t) = \Pi\left(\frac{t}{T_b}\right)$ ) has  $\infty$  bandwidth
- Band-limited signal (e.g.,  $P(f) = \Pi\left(\frac{f}{R_b}\right)$ ) needs  $\infty$  time

# INTERSYMBOL INTERFERENCE (ISI)

- Time-limited pulse transmitted over low-pass channel → Spectral distortion occurs → Pulse spreads beyond its allotted  $T_b$  interval
- Band-limited pulse transmitted over band-limited channel → Not time-limited pulse → Pulse spreads beyond its allotted  $T_b$  interval
- Either way ISI occurs!
- ISI is not noise - caused by channels that aren't distortionless over the signal bandwidth
- Pulse spreading fine **as long as no effect at *decision-making instant***

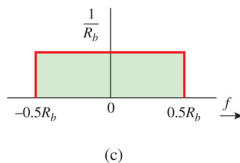
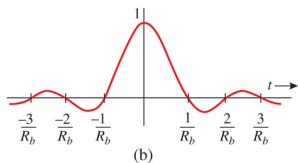
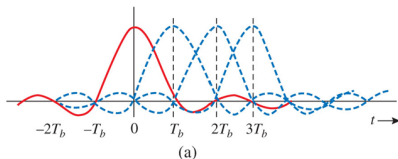
# SYMBOL DETECTION



# NYQUIST'S FIRST CRITERION FOR ZERO ISI

$$p(t) =$$

- Pulses allowed to overlap, but shaped to have zero ISI at decision points
- $p(t) = \text{sinc}(\pi R_b t)$  only one to meet Nyquist's First Criterion with min BW  $R_b/2$  Hz, but...



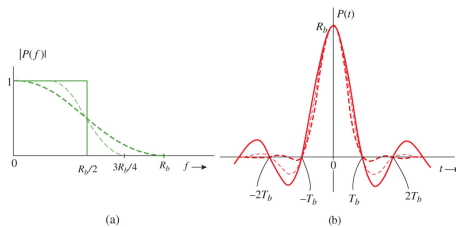
# NYQUIST'S FIRST CRITERION FOR ZERO ISI

- Want pulse with sharper roll-off to mitigate time jitter errors
- Nyquist showed this desired pulse will have bandwidth

$$B_T = \frac{(1 + r)R_b}{2}$$

where  $r$  is the **roll-off factor** between  $0 \leq r \leq 1$

- $r = \frac{\text{excess bandwidth}}{\text{theoretical minimum bandwidth}} = \frac{f_x}{R_b/2}$ , where  $f_x$  is the bandwidth in excess of min bandwidth  $R_b/2$
- Family of raised cosine pulses meets Nyquist's first criterion



# RAISED COSINE

- Where have you seen this before?
- We will come back to it in a future lesson on Matched Filters
- Lab 5 flowgraph uses Root-Raised-Cosine (RRC) filtered samples at the transmitter to control the bandwidth - it even has a parameter called "Excess Bandwidth"!
- Using the square root of the raised cosine creates ISI... but by using another RRC filter on the receiver, we end up with a regular raised cosine filter - which meets Nyquist's First Criterion and minimizes ISI!